

Software Requirement Specification (SRS)

Rear Axle Steering Control Algorithm

Requirement ID: RAS-SW-REQ-0010

1. Introduction & Scope

The Electronic Chassis Control Unit shall implement a real-time software module commanding the rear axle steering actuator to a desired angle within $\pm 5.0^\circ$ mechanical limits. It must operate in all vehicle states—low-speed maneuvers ($v < 60$ km/h), high-speed lane changes ($v > 65$ km/h), parking, reverse, and emergency maneuvers—using both rule-based open-loop and yaw-rate-feedback closed-loop control to optimize handling, stability, and safety.

2. Description & Rationale

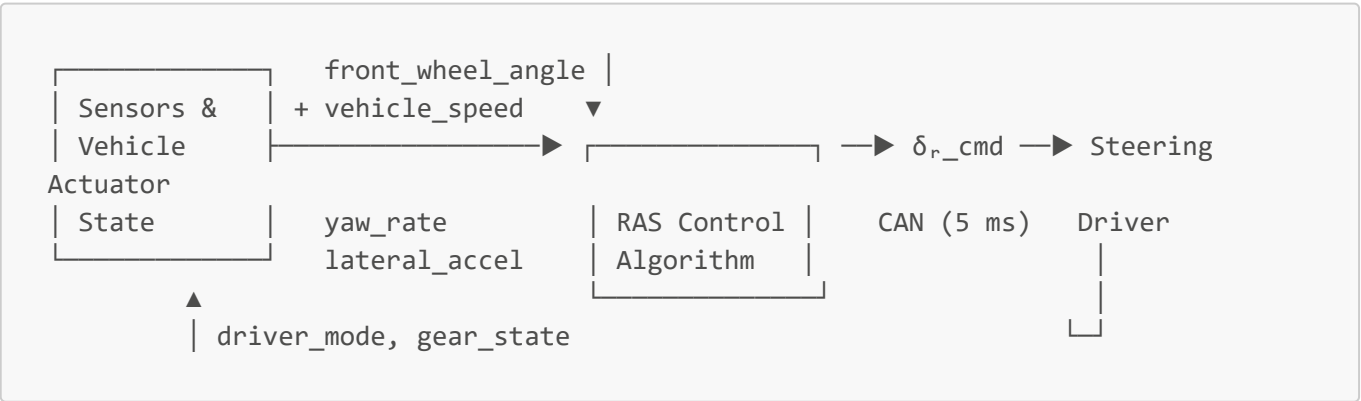
2.1 Description

- Real-time computation of δ_r (rear-wheel angle) based on front-wheel steering δ^f , vehicle speed v , yaw rate r , lateral accel a_y , driver_mode, and gear_state.
- Output δ_{r_cmd} broadcast on CAN ID 0x4B0 (0.01°/bit) every 5 ms.

2.2 Rationale

- Low-speed maneuverability:** Counter-phase steering reduces turning radius.
- High-speed stability:** Same-phase steering enhances yaw damping and lane-change control.
- Safety:** Improves obstacle-avoidance path and transient response.
- Driver confidence:** Provides predictable, seamless transitions without discontinuities.

3. System Architecture & Data Flow



Inputs:

- δ^f (front steer), v (vehicle speed), r (yaw rate), a_y (lateral accel), driver_mode, gear_state

Control Core:

1. Phase Law (rule-based)
2. PI Yaw Trim (closed-loop)
3. Hysteresis Compensation
4. Slew-Rate & Limit Enforcement

Outputs:

- δ_{r_cmd} on CAN ID 0x4B0 every 5 ms

4. Functional Requirements

4.1 Open-Loop Phase-Based Steering

4.1.1 Opposite-Phase Region ($v < 60$ km/h)

- $\delta_{ropp} = -K_1 \cdot \delta^f$
- K_1 tapers linearly from K_{1max} at 0 km/h to K_{1min} at 60 km/h (configurable via UDS).

4.1.2 Transition Blend ($55 \leq v \leq 65$ km/h)

- Weight: $w(v) = \frac{1}{1 + e^{\{-\alpha, (v-v_0)\}}}$; $\alpha=0.23$; $v_0=60$
- $\delta_{r_phase} = (1-w) \cdot \delta_{ropp} + w \cdot \delta_{rsame}$

4.1.3 Same-Phase Region ($v > 65$ km/h)

- $\delta_{rsame} = +K_2 \cdot \delta^f$
- K_2 scheduled vs. v to limit sensitivity at very high speeds.

4.2 Closed-Loop Yaw-Rate Feedback

4.2.1 Yaw-Rate Error

- $r_error = r_meas - r_model(v, \delta^f)$
- $r_model = (v / l) \cdot \tan(\delta^f) \cdot \cos(\lambda)$ (bicycle model; λ = roll-center offset)

4.2.2 PI Controller

- $\Delta\delta_r = K_p(v) \cdot r_error + K_i(v) \cdot \int r_error dt$
- K_p, K_i gain-scheduled vs. v and driver_mode; update ≥ 200 Hz; anti-windup via back-calculation.

4.3 Actuator Conditioning

4.3.1 Slew-Rate Limit

- $|d\delta_r/dt| \leq 30^\circ/s$ enforced before CAN broadcast.

4.3.2 Hysteresis Compensation

- $\delta_offset = H(\delta_cmd_last, \delta_meas) (\leq 0.3^\circ)$
- $\delta_r_cmd_adj = \delta_r_cmd + \delta_offset$

4.4 Accuracy & Fail-Safe

4.4.1 Steady-State Tolerance

- $|\delta_r\text{_{target}} - \delta_r\text{_{meas}}| \leq 0.2^\circ$ for $v \in [5, 180]$ km/h on flat road.

4.4.2 Input Invalidity

- On invalid δ_r^f , v , r , a_y , or gear_state :
 - Command $\delta_r = 0^\circ$ within 100 ms
 - Set diagnostic flag `RAS_FAIL_SAFE_ACTIVE`

5. Interface Requirements

ID	Specification
5.1	Broadcast $\delta_r\text{_{cmd}}$ on CAN ID <code>0x4B0</code> every 5 ms, scaled $0.01^\circ/\text{bit}$, offset 0° .
5.2	Required inputs: <code>front_wheel_angle</code> ($\leq 0.05^\circ$), <code>vehicle_speed</code> (≤ 0.1 km/h), <code>yaw_rate</code> ($\leq 0.2^\circ/\text{s}$), <code>lateral_accel</code> , <code>driver_mode</code> , <code>gear_state</code> .
5.3	Calibration via UDS <code>0x2F</code> (<code>WriteDataByIdentifier</code>), DataIDs <code>0xF210–0xF21F</code> .
5.4	Diagnostic routine via UDS <code>0x31</code> sub-function <code>0x0203</code> to slew actuator $\pm 3^\circ$ and verify feedback.

6. Diagnostics

ID	Specification
6.1	If $\delta_r\text{_{target}}$ $> 0.5^\circ$ for > 50 ms, store DTC <code>C1A27</code> and enter degraded mode $\delta_r\text{_{meas}}$ (δ limited to $\pm 1^\circ$).
6.2	Supervise control task with 100 ms watchdog; timeout raises <code>RAS_TIMEOUT_ERROR</code> and recenters actuator.
6.3	On power loss or ECU reset, actuator must return to 0° (mechanical centre) via spring preload within ≤ 300 ms.
6.4	Support freeze-frame capture (timestamp, v , angles, r) at fault for post-analysis.